

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 19: Aerodrome Markings & Signage

Syllabus Source: CAE Oxford (pp. 399-424) | Pass Target: >= 90% AviationExam

1. RUNWAY MARKINGS (WHITE)

Runway markings are WHITE (except yellow for displaced threshold arrows/chevrons and runway lead-in lines):

Runway Width	Number of Runway Threshold Stripes (Piano Keys)
18 m runway width	4 stripes
23 m runway width	6 stripes
30 m runway width	8 stripes
45 m runway width	12 stripes (Standard commercial jet runway)
60 m runway width	16 stripes

- Aiming Point: Two conspicuous white rectangular stripes located 400 m from threshold on runways >= 2,400 m.
- Touchdown Zone Markings: Pairs of rectangular stripes arranged symmetrically about runway centerline at 150 m intervals.
- Runway Centerline Markings: Uniformly spaced stripes and gaps: stripe length plus gap length is not less than 50 m and not more than 75 m (typically 30 m stripe and 20 m gap).
- Chevrons (Yellow): Painted on paved areas preceding threshold (blast pads, stopways) indicating area unfit for normal taxiing, take-off, or landing.

2. TAXIWAY MARKINGS & HOLDING POSITIONS (YELLOW)

- Pattern A Holding Position: Two solid yellow lines and two dashed yellow lines across taxiway. Solid lines on taxiway side; dashed lines on runway side. Aircraft MUST NOT cross solid lines towards runway without ATC clearance.
- Pattern B Holding Position: Ladder pattern ('railway tracks'). Critical area holding point for Category II / III operations. Aircraft holds here in Low Visibility Procedures (LVP) to protect ILS localizer/glidepath.
- Intermediate Holding Position: Single dashed yellow line across taxiway. Aircraft holds here until cleared by ground controller.

3. AERODROME SIGNS: MANDATORY VS INFORMATION

Sign Type	Color Scheme	Operational Meaning & Pilot Mnemonic
Mandatory Instruction Sign	WHITE letters on RED background	Identifies entrance to runway, critical area, or prohibited zone (e.g. '09-27', 'NO ENTRY', 'CAT II/III'). Must not be crossed without ATC clearance.
Location Sign	YELLOW letters on BLACK background	Identifies the taxiway or runway where the aircraft is currently located ('Black square, you are there'). Yellow border.
Direction / Destination Sign	BLACK letters on YELLOW background	Indicates designation and direction of intersecting taxiways ('Yellow lead you to the fellow'). Always includes an arrow.

Runway Exit Sign	BLACK letters on YELLOW background	Identifies runway exit taxiway (e.g. 'B ->'). Located on same side as exit.
Aerodrome Identification Sign	WHITE letters on any conspicuous color	Identifies aerodrome from air (placed on hangar roof or tower).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: How many stripes are displayed on a runway threshold marking for a 45 m wide runway?

- [A]: 8
- [B]: 12
- [C]: 16
- [D]: 6

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A 45 m runway has 12 stripes (30 m has 8; 60 m has 16). Formula:
 Stripes = Width x 4 / 15.

Q2: What visual appearance designates a mandatory instruction sign at an aerodrome?

- [A]: Yellow inscription on black background
- [B]: Black inscription on yellow background
- [C]: White inscription on red background
- [D]: White inscription on green background

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Mandatory instruction signs (e.g. STOP, NO ENTRY, runway designations) have WHITE text on a RED background.

Q3: What marking identifies an unserviceable portion of a runway or taxiway?

- [A]: A single yellow diagonal line
- [B]: A white or yellow cross (X)
- [C]: A red circle
- [D]: A dashed white box

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Unserviceable areas are marked with crosses (white on runways, yellow on taxiways).

Q4: What color is used for taxiway centerline markings?

- [A]: White
- [B]: Yellow
- [C]: Red
- [D]: Blue

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. All taxiway markings, holding lines, and guidance lines are YELLOW (runway markings are white).

Q5: What visual appearance designates a location sign (identifying the taxiway or runway on which the aircraft

- [A]: Yellow text on black background with yellow border
- [B]: Black text on yellow background
- [C]: White text on red background
- [D]: White text on green background

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Location signs display YELLOW text on a BLACK background with a yellow border ('Yellow on black is where you're at; Black on yellow is where to go').

Q6: What is the meaning of a runway-holding position marking consisting of two continuous yellow lines and t

- [A]: Intermediate holding position on apron
- [B]: Mandatory runway-holding position: when approaching from continuous lines, MUST NOT cross without ATC clearance
- [C]: Road-holding position for ground vehicles
- [D]: De-icing entry line

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Pattern 'A' runway-holding position consists of two continuous lines on taxiway side and two dashed lines on runway side. An aircraft approaching from the continuous side must stop.